



RKDF University, Gandhinagar, Bhopal

Faculty Profile

Basic Information				
Name		DR RITESH SADIWALA		
Date of Birth		08.12.1978		
Designation		PROFESSOR AND HEAD		
Department		ELECTRONICS AND COMMUNICATION ENGINEERING		
Experience		22 YEARS		
Email ID		ritesh14ci@gmail.com		
Contact No		7049589573		
Educational Qualification				
Description	Subject	Year	Percentage	Institute/ University
UG	Electronics and Communication Engineering	2001	75.82	Rajiv Gandhi Proudtyogiki Vishwavidyalaya , Bhopal (G.E.C. Bhopal) at present UIT, Bhopal
PG	Digital Communication	2006	88	Maulana Azad National Institute of Technology, Bhopal
Ph.D.	Electronics and Communication Engineering	2017		RKDF University, Gandhinagar, Bhopal
Post Doctorate				
NET Qualified				
Experience Detail				
Experience (Teaching/Research)	Designation	Duration		Name of Institute/University
		From	To	
Industry	Executive Engineer	Oct’05	Dec’08	Indian Telephone Industry (A Government of India Undertaking)
Industry	Senior Engineer Maintenance –‘NS	Dec’08	Dec’11	Indian Telephone Industry (A Government of India Undertaking)
Industry	Manager Project –‘NS’	Dec’11	July’16	Indian Telephone Industry (A Government of India Undertaking)
Industry	Senior Technical Leader	July 2016	October 2017	4 Consulting Inc. Dallas/Fort Worth Area, Texas
Teaching	Assistant Professor	Dec’2017	Oct’2021	RKDF University, Gandhinagar, Bhopal
Teaching	Vice Principal	Oct’2021	May 2023	RKDF University, Gandhinagar, Bhopal
Teaching	Professor and Head	May 2023	Continuing	RKDF University, Gandhinagar, Bhopal

Publications			
No. of Research Papers Published(Attach Annexure of Research papers published)	60 (Annexure A attached)		
No.of Books Published	1 (Annexure B attached)		
Books Chapters Published(Provide print of 1 st Page of Book)	4 (Annexure B attached)		
Research Scholars Supervised			
Research Program	Award	Under Supevision	Name of the University
Ph.D. (Annexure C attached)	9	1	RKDF UNIVERSITY,GANDHINAGAR,BHOPAL
M.Phil			
PG Thesis /Dissertation	12	2	RKDF UNIVERSITY,GANDHINAGAR,BHOPAL
Area of Expertise(100 words)			
My field is Electronics and Communication Engineering. My area of expertise consists of communication systems and in that particular wireless communication, optical communication and Network communication. I am learning and specializing in data transmission over computer networks, including protocols, network design, and the management of local and wide area networks (LANs and WANs) and trying to update my knowledge in it. I am also specialized in Transistor Design and Fabrication and in the development and deployment of sensor networks that collect data and have expertise in designing energy-efficient circuits and communication systems that are suitable for low-power wireless sensor applications.			
Award and Achievement			
Name of Award	Description(With certified copy of the award)		
National			
International			
Conference/Seminar/Workshops/FDP			
Description			No.
Conference Attended			19
Conference Organized			4
Seminar Attended			7
Seminar Organized			2
Workshop Attended			9
Workshop Organized			2
FDP Attended			2
FDP Organized			2
Research Project			
Name of Project	Funding Agencies	Amount	
Any other Achievement			
Patents Granted/Published	Title of the Patent	Year	
Patents Granted	Smart Clinical Site Selecting Robotic Device	2024	
	Smart Clinical Site Selecting Robotic Device	2023	
	IOT Based Patient Monitoring System Using Sensors to Detect, Analyse And Monitor Two Primary Vital Signs	2022	
	Secure and Dynamic Key Management In Wireless Sensor Networks	2022	
Patents Published	Method for Migration of Sessions Among Cipher Machines Using Lagrange Interpolation	2022	
	Touch Screen Devices Employing Graphene Electrodes Field of Invention Computer Science	2021	

Signature

ANNEXURE A

ANNEXURE A

Research Publications (60)

(A) International Journal Publications (40)

1. Ambresh Patel and Dr. Ritesh Sadiwala, "Performance Analysis of Various CMOS Logics for High Speed VLSI Circuits", Samriddhi –JPSET, Volume 15(1), Jan 2023. **(UGC Care Listed)**
<https://doi.org/10.18090/10.18090/samriddhi.v15i01.13>
2. Ambresh Patel and Dr. Ritesh Sadiwala, "Optimizing and Recuperating the Leakages in Low Voltage CMOS Circuits" Samriddhi –JPSET, Volume 14, Issue 2, 2022, pp 1-4. **(UGC Care Listed)**
<https://doi.org/10.18090/samriddhi.v14i02.13>
3. Chandan Kumar Roy and Dr. Ritesh Sadiwala, "Smart Environment for IOT based Disease Diagnosis Healthcare Framework", YMER, Volume 21, Issue 11, Nov 2022, pg 446-458. **[Indexed in SCOPUS]**
<https://pubs.thesciencein.org/journal/index.php/jist/article/view/a731>
4. Jayant Dorave and Dr. Ritesh Sadiwala, "Numerous Pursuer Drones Immobilizing Multiple Evader Drones by Switching Target Enabled Communication Method in Smart Dust-IoT Environment", Design Engineering, 2021, pp 9735-9748. **[Indexed in SCOPUS]**
5. Jayant Dorave and Dr. Ritesh Sadiwala, "A Secure Communication Protocol for Unmanned Aerial Vehicles Using IOT Protocols", Samriddhi –JPSET, Volume 14(4), Oct 2022. **(UGC Care Listed)**
DOI: [10.18090/samriddhi.v14i04.13](https://doi.org/10.18090/samriddhi.v14i04.13)
6. Jayant Hande and Dr. Ritesh Sadiwala, "Data Security-Based Routing in MANETs Using Key Management Mechanism" SN Computer Science, Jan 2024, SN Computer Science, <https://doi.org/10.1007/s42979-023-02409-5> **[Indexed in SCOPUS]**
7. Jaishri Tiwari and Dr. Ritesh Sadiwala, "Personality prediction from Five-Factor Facial Traits using Deep learning" Journal of Integrated Science and Technology: Vol. 11 No. 4 (2023) **[Indexed in SCOPUS]**
<https://pubs.thesciencein.org/journal/index.php/jist/article/view/a578>
8. Hemant Rajoriya, Ritesh Sadiwala, "Deep compressive sensing and reconstruction algorithm in wireless Internet of Things", Journal of Integrated Science and Technology: Vol. 11 No. 2 (2023) **[Indexed in SCOPUS]**
<https://pubs.thesciencein.org/journal/index.php/jist/article/view/487>
9. Kuldeep Pandey, Ritesh Sadiwala, "Dual-Port MIMO antenna design for IoT: Analysis and Implementation, Journal of Integrated Science and Technology", Vol. 12 No. 3 (2024) **[Indexed in SCOPUS]**
<https://doi.org/10.62110/sciencein.jist.2024.v12.768>
10. Ajay Kumar Barapatre, Ritesh Sadiwala, "Energy and resources management for Multiple Access in Massive IOT network", Journal of Integrated Science and Technology: Vol. 11 No. 3 (2023) **[Indexed in SCOPUS]**
<https://pubs.thesciencein.org/journal/index.php/jist/article/view/a518>
11. Jayant Y. Hande, Ritesh Sadiwala, "Optimization of energy consumption and routing in MANET using Artificial Neural Network , Journal of Integrated Science and Technology", Vol. 12 No. 1 (2024) **[Indexed in SCOPUS]**
12. Chandan Kumar Roy, Ritesh Sadiwala, "Smart internet of things (IoT) based healthcare framework environment for Chikungunya disease diagnosis", Journal of Integrated Science and Technology: Vol. 12 No. 4 (2024) **[Indexed in SCOPUS]** <https://doi.org/10.62110/sciencein.jist.2024.v12.779>
13. Chandan Kumar Roy, Ritesh Sadiwala, "Smart environment for IoT based disease diagnosis healthcare framework", Journal of Integrated Science and Technology: Vol. 12 No. 2 (2024) **[Indexed in SCOPUS]**

<https://pubs.thesciencein.org/journal/index.php/jist/article/view/a731>

14. Kuldeep Pandey, Ritesh Sadiwala, "Performance evaluation of FR2-based IoT and AI applications using a Quad Port Multiple Input Multiple Output Antenna", Journal of Integrated Science and Technology: Vol. 11 No. 4 (2023) **[Indexed in SCOPUS]**

<https://pubs.thesciencein.org/journal/index.php/jist/article/view/a567>

15. Ajay K Barapatre, Ritesh Sadiwala, "Resource Allocation in Massive Internet of Things-Edge Network with Optimal Path Planning and Scheduling", Samriddhi: A Journal of Physical Sciences, Engineering and Technology, Volume 14, Issue 4, 2022, Print ISSN: 2229-7111, Online ISSN: 2454-5767, [Doi: 10.18090/samriddhi.v14i04.04](https://doi.org/10.18090/samriddhi.v14i04.04) **(UGC Care Listed)**
16. Jaishri Tiwari and Dr. Ritesh Sadiwala, "Personality Prediction from Handwriting using Fine-tuned Transfer Learning Models", SAMRIDDHI, ISSN: 2454-5767, Volume 15, Issue 1, 2023 **(UGC Care Listed)**
<https://doi.org/10.18090/samriddhi.v15i01.06>
17. Kuldeep Pandey and Dr. Ritesh Sadiwala, "Parametric Analysis of Circular Patch Wideband Antenna for IOT Application", Lecture Notes in Electrical Engineering (Advances in VLSI, Communication, and Signal Processing), Oct 5-6 2021 & IIT BHU Varanasi. **[Indexed in SCOPUS]**
https://www.researchgate.net/publication/364160146_Parametric_Analysis_of_Circular_Patch_Wideband_Antenna_for_IoT_Application
18. Hemant Rajoriya and Dr. Ritesh Sadiwala, "Deep compressive sensing and reconstruction algorithm in wireless Internet of Things" Journal of Integrated Science & Technology, ISSN: 2321-4635, J. Integr. Sci. Technol., 2023, 11(2), 487 URN:NBN: sciencein. jist.2023.v11.487. **[Indexed in SCOPUS]**
<https://pubs.thesciencein.org/journal/index.php/jist/article/view/487>
19. Hemant Rajoriya and Dr. Ritesh Sadiwala, "Data Traffic Reduction in Wireless Internet of Things Using Deep Compressive Sensing and Reconstruction Algorithm" A Journal of Physical Sciences, Engineering and Technology, Volume 15, Issue 1 (2023) Print ISSN: 2229-7111 Online ISSN: 2454-5767. **(UGC Care Listed)** DOI: [10.18090/samriddhi.v15i01.32](https://doi.org/10.18090/samriddhi.v15i01.32)
20. Jayant Y. Hande, and Ritesh Sadiwala, "Optimization of energy consumption and routing in MANET using Artificial Neural Network", Journal of Integrated Science & Technology Cite as: J. Integr. Sci. Technol., 2024, 12(1), 718. Published on: 11-Sep-2023 **(UGC Care Listed)**
<https://pubs.thesciencein.org/journal/index.php/jist/article/view/a718>
21. Jayant Y. Hande and Dr. Ritesh Sadiwala, "A Secure Adaptive Bacterial Foraging Optimization Routing Algorithm for Mobile Ad Hoc Networks", TELEMATIQUE, Volume 21 Issue 1, 2022 **[Indexed in SCOPUS]**
22. Farooq Fareed Nabi, Dr Ritesh Sadiwala and Dr Ratnesh Jain, "Empirical Study: Status of Management of Microwave Radiation Hazards in the Government Organisation" Juni Khyat ISSN: 2278-4632, Vol-14, Issue-4, No.04, April 2024 **(UGC Care Listed)**
23. Jayant Dorave and Dr. Ritesh Sadiwala, "IOT and Smart Dust Architecture Integration for Network Optimization using Artificial Neural Network", International Journal for Innovative Engineering and Management Research(IJIEMR), Vol- 10, Issue 11, Nov 2021. DOI: [10.48047/IJIEMR/V10/11/13](https://doi.org/10.48047/IJIEMR/V10/11/13)
24. Jayant kumar Dorave and Dr. Ritesh Sadiwala, "Uplink and Downlink Variation in Drone Technology for Cloud, Edge, Fog and Smart Dust Integrated IoT Architecture: Demonstrated Over WSNs", AMSE 2021, Journal of Physics: Conference Series, IOP Publishing, [doi:10.1088/1742-6596/2089/1/012023](https://doi.org/10.1088/1742-6596/2089/1/012023).
25. Nadia Sharif and Dr. Ritesh Sadiwala, "A Methodical and Erudite Study on 5G Wireless Networks", SHODH SANGAM -- A RKDF University Journal of Science and Engineering (ISSN No. 2581-5806), March 2020, pp 52-61.
26. Nadia Sharif and Dr. Ritesh Sadiwala, "Performance Analysis of QOS Parameters of 5G Wireless Communication Networks", Research Journal of Engineering Technology and Management (ISSN: 2582-0028), Volume 03, Issue 02, June-2020, pp 66-75.
27. SanjerJilani and Dr. Ritesh Sadiwala, "A Methodical and Erudite Study on 5G Wireless Networks", SHODH SANGAM (ISSN No. 2581-5806), A RKDF University Journal of Science and Engineering (ISSN No. 2581-5806), Vol.-04, No.-02, April-2021, pp 34-40.
28. SanjerJilani and Dr. Ritesh Sadiwala, "Performance Analysis of QOS Parameters of 5G Wireless Communication Networks", Research Journal of Engineering Technology and Medical Sciences (ISSN: 2582-6212), Special Issue 2

(April 2020), pp 97-109

29. Rajnish Mishra and Dr. Ritesh Sadiwala, "Triangular Potential Well Approximation in Scaled MOSFET for Low Frequency Noise Modeling and Characterization". Research Journal of Engineering Technology and Medical Sciences (ISSN: 2582-6212), Volume 03, Issue 03, Sep-2020, pp 41-51.
30. Rajnish Mishra and Dr. Ritesh Sadiwala, "Survey on Modelling and Characterization of MOSFET for Low Frequency Noise", SHODH SANGAM -- A RKDF University Journal of Science and Engineering (ISSN No. 2581-5806), Vol-03, No.06, Nov 2020, pp 109-115.
31. Enamur Rahman Shamsi and Dr. Ritesh Sadiwala, "Energy Estimated Mobile Sink based Routing Protocol in Wireless Sensor Network," International Journal of Research & Technology (IJRT), Dec 2018, Vol 06, Issue 06, pp. 25-28.
32. Jayant D Dorave and Dr. Ritesh Sadiwala, "Compatibility and Integration of Smart Dust with various frequencies of electromagnetic spectrum and physical systems", The Splendid Journals, June 2019, Vol 2, Issue 3, pp 1-6.
33. Enamur Rahman Shamsi and Dr. Ritesh Sadiwala, "Survey of Clustering based Application Specific Routing Protocols in Wireless Sensor Network," SHODH SANGAM – A RKDF University Journal of Science and Engineering, (ISSN No. 2581-5806), Feb. 2019, Vol-2, Issue-3, pp. 55-58.
34. Ritesh Sadiwala and Dr. Minal Saxena, "Performance Evaluation of Quality Parameters in VOIP and PSTN Systems" IRACST – International Journal of Computer Networks and Wireless Communications (IJCNCW), ISSN: 2250-3501 Vol.5, No 5, October 2015, pp 621-627
35. Ritesh Sadiwala and Dr. Minal Saxena "Performance Evaluation of Next Generation Networks using OPNET Simulator" Communications on Applied Electronics (CAE) – ISSN: 2394-4714 Foundation of Computer Science FCS, New York, USA Volume 2 – No.4, July 2015, pp 32-37
36. Ritesh Sadiwala and Dr. Minal Saxena, "Transformation and Analysis of Wire line Networks to Wireless Networks Using OPNET Simulator" International Journal of Broadband Cellular Communication Vol. 1: Issue 1, July 2015, pp 20-35
37. Ritesh Sadiwala and Dr. Manisha Sharma "SECURITY THREATS OF VOIP" Journal of Innovative trends in Science, Pharmacy & Technology, Issue 1, Vol-1(1), August 2014, pp 7-18.
38. Ritesh Sadiwala and Dr. Manisha Sharma "PERFORMANCE ANALYSIS OF VOIP NETWORKS USING NETWORK SIMULATORS-A REVIEW" International Journal of Computer Networking and Communication (IJCNC), Vol. 2, No. 3 (August 2014), pp 1-15.
39. Ritesh C Sadiwala "Convergence towards Next Generation Wireless Networks" IOSR Journal of Electrical and Electronics Engineering (IOSR-JEEE) e-ISSN: 2278-1676 Volume 4, Issue 4 (Jan. - Feb. 2013), pp 01-13.
40. Ritesh C Sadiwala "Energy Efficiency for Long Term Evolution Cellular Networks", International Journal of Engineering Research & Technology (IJERT) ISSN: 2278-0181 Vol. 2 Issue 10, October – 2013, pp 1968-1982.

(B) Conference Publications (20)

1. Ritesh Sadiwala "Performance Evaluation of Quality Parameters for Integrated Unified Communication Network", 6 th International Conference on "Computing for Sustainable Global Development", 13th - 15th March, 2019 Bharati Vidyapeeth's Institute of Computer Applications and Management (BVICAM), INDIA Com-2019, New Delhi (INDIA) IEEE Conference ID: 46181 [Indexed in SCOPUS]
2. Braj Kishor, Dr. Ritesh Sadiwala, "Dynamic CMOS Logic using clock gating in Deep Submicron Technology" 2nd International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, Shodh Sangam

September 28-29, 2019.

3. BrajKishor, Dr. Ritesh Sadiwala "Performance analysis of Dynamic CMOS Logic using Submicron Technology" 3rd International Conference on Recent Advances in smart Innovative Ideas with Multidisciplinary Research, Nov 30-Dec01, 2019.
4. Thirupathi Durgam and Dr. Ritesh Sadiwala, "Dynamic time assisted authentication protocol (DTAAP) for client-server in WSN-IoT environment", 2021 International Conference on Emerging Smart Computing and Informatics, ESCI 2021, 2021, pp. 583–586, ID no. 9397039 **[Indexed in IEEE Xplore, SCOPUS]**
[DOI:10.1109/ESCI50559.2021.9397039](https://doi.org/10.1109/ESCI50559.2021.9397039)
5. Chandan Kumar Roy and Dr. Ritesh Sadiwala "Cloud-fog based healthcare framework to identify and prevent dengue fever outbreak", 2021 International Conference on Emerging Smart Computing and Informatics, ESCI 2021, 2021, pp. 537–541, Id no. 9396905 **[Indexed in IEEE Xplore, SCOPUS]** [DOI:10.1109/ESCI50559.2021.9396905](https://doi.org/10.1109/ESCI50559.2021.9396905)
6. Ambresh Patel and Dr. Ritesh Sadiwala "Low voltage cmos switch capacitor circuit", Proceedings of the 7th International Conference on Electrical Energy Systems, ICEES 2021, 2021, pp. 629–634, Id no. 9383686 **[Indexed in IEEE Xplore, SCOPUS]** [DOI:10.1109/ICEES51510.2021.9383686](https://doi.org/10.1109/ICEES51510.2021.9383686)
7. Chandan Kumar Roy and Dr. Ritesh Sadiwala "An Intelligent Fog-IoT based Disease Diagnosis Healthcare System", 4th International Conference on Soft Computing and Signal Processing (ICSCSP-2021), Proceedings of ICSCSP-2021, Springer- Advances in Intelligent Systems and Computing (AISC) Series. **[Indexed in IEEE Xplore, SCOPUS]**
[DOI:10.1007/978-981-16-7088-6_33](https://doi.org/10.1007/978-981-16-7088-6_33)
8. Thirupathi Durgam and Dr. Ritesh Sadiwala, "An Efficient Group Key Establishment for Secure Communication to multicast groups for WSN-IoT nodes", 4th International Conference on Soft Computing and Signal Processing (ICSCSP-2021), Proceedings of ICSCSP-2021, Springer- Advances in Intelligent Systems and Computing (AISC) Series. **[Indexed in IEEE Xplore, SCOPUS]** [DOI:10.1007/978-981-16-7088-6_64](https://doi.org/10.1007/978-981-16-7088-6_64)
9. Jaishri Tiwari and Dr. Ritesh Sadiwala, "Analysis of Personality Traits of Handwriting with Comparison of Different Techniques of Personality Detection" 2022 IEEE International Conference on Current Development in Engineering and Technology (CCET), 23-24 December 2022, [DOI: 10.1109/CCET56606.2022.10080717](https://doi.org/10.1109/CCET56606.2022.10080717) **[Indexed in IEEE Xplore, SCOPUS]**
10. Ambresh Patel and Dr. Ritesh Sadiwala, "Low Voltage CMOS Switch Capacitor Circuit, 2021 7th International Conference on Electrical Energy Systems (ICEES), 11-13 February 2021, [DOI:10.1109/ICEES51510.2021.9383686](https://doi.org/10.1109/ICEES51510.2021.9383686) **[Indexed in IEEE Xplore, SCOPUS]**
11. Ajay Kumar Barapatre, Ritesh Sadiwala, "A Study on Resource Allocation for Edge Computing in IOT Networks", International E-Conference on Impact of Science, Engineering and Technology Advancements in Humanities, Management, Medical and Nursing Sciences" on 19 Nov 2022.
12. Ajay Kumar Barapatre, Ritesh Sadiwala, "Collision Avoidance for Multiple Access in Massive IOT", MPCST sponsored National Conference on Contemporary Technological Solutions towards fulfillment of Social Needs at RKDF University, Gandhi Nagar, Bhopal on 03-04 Nov 2023.
13. Jaishri Tiwari and Dr. Ritesh Sadiwala, "Personality Detection with AI using Deep learning approach", International Conference on Recent Advances in Science and Engineering, SAGE University, Bhopal, April 23, 2022
14. Jayant Y. Hande and Dr. Ritesh Sadiwala, "Routing Protocols applied in disaster management for future research using MANET: A review", National Conference - Advances in Science, Engineering & Technology for Sustainable Development Goals, held on Dec. 06-08, 2022 at Madhyanchal Professional University, Bhopal (M.P.)
15. Jayant Y. Hande and Dr. Ritesh Sadiwala, "Study of various Routing Protocols for MANETs: - A prospect review study", International Conference on Futuristic Research in Modeling of Dynamical Systems, (ICFRMDS-2022) held on 4th and 5th of May 2022, at CMR INSTITUTE OF TECHNOLOGY, BENGALURU.
16. Jayant Y. Hande and Dr. Ritesh Sadiwala, "Comparative study of Various Optimization techniques in MANET", International Conference on Advance research Tools Technique for Multidisciplinary Research design organised by Eudoxia Research University, USA and Eudoxia Research Centre, India held on 9th and 10th of December 2022.
17. Jayant Dorave and Dr. Ritesh Sadiwala, "Uplink and Downlink Variation in Drone Technology for Cloud, Edge, Fog and Smart Dust Integrated IoT Architecture: Demonstrated over WSNs", 1st International Conference on Applied Mathematics, Modeling and Simulation in Engineering (AMSE 2021), Stanley College of Engineering and

Technology, Hyderabad, India held on 15-16 September 2021.

- 18.Kuldeep Pandey and Dr. Ritesh Sadiwala, “Dual Port Multiple Input Multiple Output Antenna for 5G Communication Applications”,in the 2nd International Conference on Advances in Signal Processing and Communication Engineering (ICASPACE-2023) **[Indexed in IEEE Xplore, SCOPUS]**
- 19.Jayant Dorave and Dr. Ritesh Sadiwala, “ IOT and Smart Dust Architecture Integration for Network Optimization using Artificial Neural Network”, International Journal for Innovative Engineering and Management Research(IJIEMR), Vol- 10, Issue 11, Nov 2021**[Indexed in IEEE Xplore, SCOPUS]**
- 20.Hande, J.Y., Sadiwala, Ritesh. Data Security-Based Routing in MANETs Using Key Management Mechanism. SN COMPUT. SCI. 5, 155 (2024). **[Indexed in SCOPUS]**

7. Paper Presentation International/ National Conferences(06)

- 1) Ritesh Sadiwala “Performance Evaluation of Quality Parameters for Integrated Unified Communication Network”, 6 th International Conference on “Computing for Sustainable Global Development”, 13th - 15th March, 2019 Bharati Vidyapeeth's Institute of Computer Applications and Management (BVICAM), INDIA Com-2019, New Delhi (INDIA) IEEE Conference ID: 46181[**Indexed in SCOPUS**]
- 2) BrajKishor, Dr. Ritesh Sadiwala, “Dynamic CMOS Logic using clock gating in Deep Submicron Technology” 2nd International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, ShodhSangam September 28-29, 2019.
- 3) BrajKishor, Dr. Ritesh Sadiwala “Performance analysis of Dynamic CMOS Logic using Submicron Technology” 3rd International Conference on Recent Advances in smart Innovative Ideas with Multidisciplinary Research, Nov 30- Dec 01, 2019.
- 4) Thirupathi Durgam and Dr. Ritesh Sadiwala, ”Dynamic time assisted authentication protocol (DTAAP) for client-server in WSN-IoT environment”, 2021 International Conference on Emerging Smart Computing and Informatics, ESCI 2021, 2021, pp. 583–586, ID no. 9397039[**Indexed in IEEE Xplore, SCOPUS**]
- 5) Thirupathi Durgam and Dr. Ritesh Sadiwala, “An Efficient Group Key Establishment for Secure Communication to multicast groups for WSN-IoT nodes “, 4th International Conference on Soft Computing and Signal Processing (ICSCSP-2021), Proceedings of ICSCSP-2021, Springer- Advances in Intelligent Systems and Computing (AISC) Series. [**Indexed in IEEE Xplore, SCOPUS**]
- 6) Dr. Ritesh Sadiwala, “Dynamic CMOS Logic using clock gating in Deep Submicron Technology” 2nd International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, Shodh Sangam September 28-29, 2019.

ANNEXURE B

(A) Book Published:-

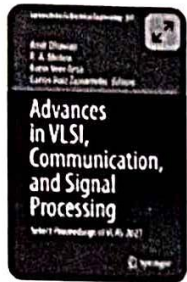
S. No.	Title of Book/Monograph	Authors	Editor	Year	ISBN No.	Publisher Name
1.	Improvement Measures of Key Parameters for Future Wireless Networks	Dr. Ritesh Sadiwala	Ecaterina Focsa	2022	9786202300377	Scholars' Press

(B) BookChapterPublished (4)

- 1) Pandey, K., Shankhwar, A.K., Sadiwala, Ritesh (2022). Parametric Analysis of Circular Patch Wideband Antenna for IoT Application. In: Dhawan, A., Mishra, R.A., Arya, K.V., Zamarreño, C.R. (eds) Advances in VLSI, Communication, and Signal Processing. Lecture Notes in Electrical Engineering, vol 911. Springer, Singapore. https://doi.org/10.1007/978-981-19-2631-0_9 [Indexed in SCOPUS]
- 2) Durgam, T., Sadiwala, R. (2022). An Efficient Group Key Establishment for Secure Communication to Multicast Groups for WSN-IoT Nodes. In: Reddy, V.S., Prasad, V.K., Wang, J., Reddy, K. (eds) Soft Computing and Signal Processing. ICSCSP 2021. Advances in Intelligent Systems and Computing, vol 1413. Springer, Singapore. https://doi.org/10.1007/978-981-16-7088-6_64 [Indexed in SCOPUS]
- 3) Roy, C.K., Sadiwala, R. (2022). An Intelligent Fog-IoT-Based Disease Diagnosis Healthcare System. In: Reddy, V.S., Prasad, V.K., Wang, J., Reddy, K. (eds) Soft Computing and Signal Processing. ICSCSP 2021. Advances in Intelligent Systems and Computing, vol 1413. Springer, Singapore. https://doi.org/10.1007/978-981-16-7088-6_33 [Indexed in SCOPUS]
- 4) Pandey, K., Pandey, R., Sadiwala, R. (2024). Dual Port Multiple Input Multiple Output Antenna for 5G Communication Applications. In: Kumar Jain, P., Nath Singh, Y., Gollapalli, R.P., Singh, S.P. (eds) Advances in Signal Processing and Communication Engineering. ICASPACE 2023. Lecture Notes in Electrical Engineering, vol 1157. Springer, Singapore. https://doi.org/10.1007/978-981-97-0562-7_36 [Indexed in SCOPUS]

Conference Proceedings (2)

1. Dr. Ritesh Sadiwala (2019) "Performance Evaluation of Quality Parameters for Integrated Unified Communication Network", IEEE Explorer, <https://ieeexplore.ieee.org/document/8991319> [Indexed in IEEE Explorer]
2. Ambresh Patel , Dr. Ritesh Sadiwala (2021) "Low Voltage CMOS Switch Capacitor Circuit" ,IEEE Explorer , <https://www.proceedings.com/content/058/058377webtoc.pdf> [Indexed in IEEE Explorer]

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| Conference paper | First Online: 05 October 2022

| pp 97–105 | [Cite this conference paper](#)



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 682 Accesses

Abstract

In this paper, a novel topology microstrip dual-side printed antenna is presented for IoT-enabled applications. The novel structured antenna comprises a circular patch by cutting off the ground plane. The ground plane along with a small rectangular edge is exactly below the feed line to increase the antenna bandwidth and a circular notch on the surface of the ground plane is used to increase the antenna properties. Therewith, multiple rectangular patches in the ground plane have increased the effective area of the antenna. The proposed antenna covers the wideband frequency range of 1.15 to 4.83 GHz

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An Efficient Group Key Establishment for Secure Communication to Multicast Groups for WSN-IoT Nodes

| Conference paper | First Online: 15 February 2022

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[Soft Computing and Signal Processing](#)
(ICSCSP 2021)

[Thirupathi Durgam](#) & [Ritesh Sadiwala](#)

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Abstract

Wireless sensor networks (WSNs) are an important element for Internet of Things technologies (IoT). The radio, multimedia, group communications deliver effective way

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| Conference paper | First Online: 15 February 2022

| pp 359–370 | [Cite this conference paper](#)



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(ICSCSP 2021)

[Chandan Kumar Roy & Ritesh Sadiwala](#)

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Abstract

Modern health services are the greatest problem in developed countries in particular, where remote regions are not supplied with good quality drugs and hospitals. IoT is a major player in medical treatment to provide people with better clinical services, which

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Dual Port Multiple Input Multiple Output Antenna for 5G Communication Applications

| Conference paper | First Online: 04 July 2024

| pp 473–479 | [Cite this conference paper](#)



[Advances in Signal Processing and
Communication Engineering](#)
(ICASPACE 2023)

[Kuldeep Pandey](#) , [Rashmi Pandey](#) & [Ritesh Sadiwala](#)

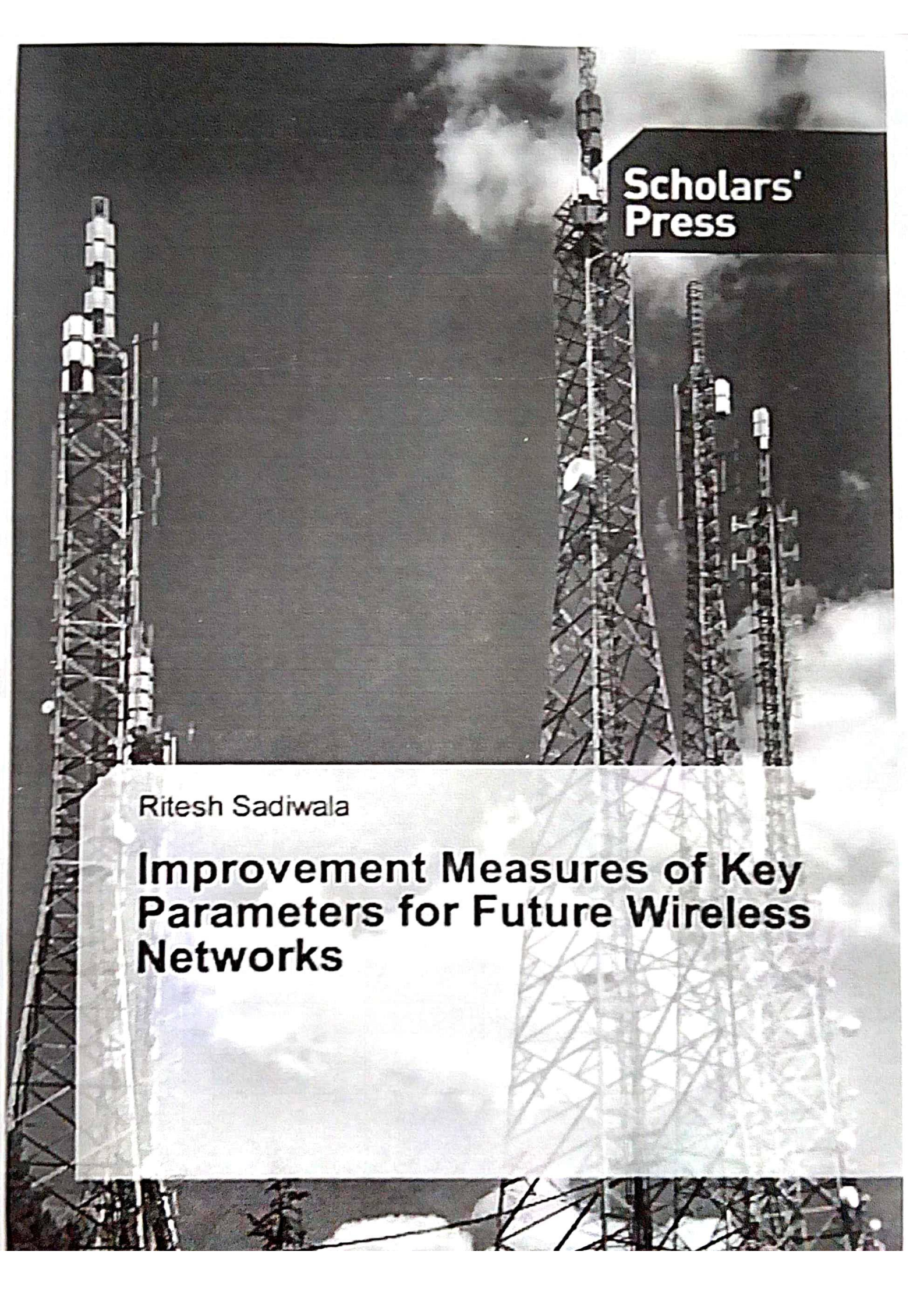
 Part of the book series: [Lecture Notes in Electrical Engineering](#) ((LNEE, volume 1157))

 Included in the following conference series:
[International Conference on Advances in Signal Processing And Communication Engineering](#)

 114 Accesses

Abstract

This paper presents the design of a two-element MIMO antenna for 5G applications. The dimensions of the proposed MIMO antenna remain at $30 \text{ mm} \times 30 \text{ mm}^2$, and it is constructed from an FR-4 substrate with a 1.6 mm thickness. A defective ground



**Scholars'
Press**

Ritesh Sadiwala

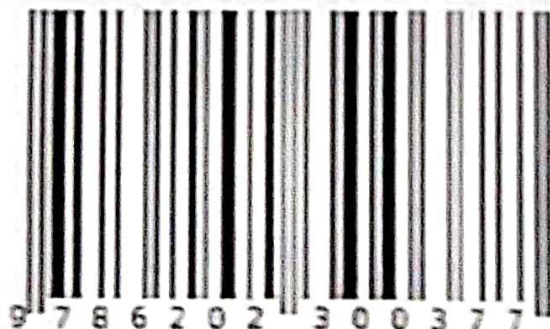
Improvement Measures of Key Parameters for Future Wireless Networks

Improvement Measures of Key Parameters for Future Wireless Networks

The enormous success of the Internet has led to a great revolution in Telecommunication. Today, Telecommunication has become an indispensable part of everybody's day-to-day activities. Various types of services like online video, online music, fast video streaming, video conferencing, VoIP, etc. are becoming a part of daily and routine activities. These applications demand high quality and congestion free networks particularly for voice and real time sessions. This can be possible when the key focus for telecommunication area will be on the Next Generation Wireless Networks (NGWNs). The idea behind this book is to Judge the key parameters of Next generation wireless networks so as to get a good network quality and a congestion free next generation network. The aim is achieved by analyzing various quality parameters like Jitter, Mean opinion Score(MOS), End to End Delay, etc. of the future networks. Next Generation Systems are advanced, fast and has several applications. The suggestions given in this book when implemented will definitely improve the performance of the Future Generation Networks by 90-95%.



Dr Ritesh Sadiwala has done his M.Tech from MANIT Bhopal (M.P.) India. He has done his Ph.D in Electronics and Communication Engineering . His areas of interest are Next Generation Networks, Wireless Sensor Networks .He has fifteen years of Industry and Eleven years of Teaching Experience. He has published many research papers in various Journals.



ANNEXURE C

Ph.D. Research Scholars Guidance

Status	Number
Completed	09
On-going	01

Details of the Ph.D Completed

S.NO.	NAME OF THE STUDENT	TITLE OF THESIS	YEAR OF AWARD
1.	AJAY KUMAR BARAPATRE	Resource Management for Multiple Access in Massive IoT Network	2024
2.	JAISHRI TIWARI	Image based Personality Prediction using Hybrid Deep Learning Approach	2024
3.	JAYANT HANDE	Optimization of energy consumption and routing in MANET using Artificial Neural Network	2024
4.	HEMANT RAJORIA	Data Traffic Reduction in Wireless Internet of Things Using Deep Compressive Sensing and Reconstruction Algorithm	2024
5.	KULDEEP PANDEY	Design and Performance Analysis of Multiple Input Multiple Output(MIMO) Smart Antenna for 5G	2024
6.	AMBRESH PATEL	Performance Analysis and Improvement Measures of Low Leakage CMOS Switch for Low Voltage Circuits	2024
7.	CHANDAN KUMAR ROY	Monitoring and Diagnosis of Some Human Diseases using IoT Systems	2025
8.	FAROOQ FAREED NABI	Microwave Radiation Induced Hazards Management in Government Organization Systems	2025
9.	JAYANTKUMAR DHANRAJ DORAVE	Analysis and Improvement of Performance Indicators at 2.4 GHz and 5GHz Network Communication Systems using Smart Dust Mote Sensor Networks	2025

